

AFFILIATIONS & EDUCATION

University of Oxford Florence Nightingale Bicentenary Fellow	Oxford, UK 2026–Present
University College London PhD in Statistical Science	London, UK 2022–2026
– Thesis: Reconciling Robustness and Tractability in Bayesian Inference under Data Contamination – Supervisors: Dr. Jeremias Knoblauch, Prof. François-Xavier Briol	
Universidad de Chile M.S. in Engineering Sciences, Applied Mathematics	Santiago, Chile 2020–2021
– Thesis: “Nonstationary multi-output Gaussian processes via harmonizable spectral mixtures” – Supervisor: Dr. Felipe Tobar – Ranked in top 10%. Best thesis award	
École Centrale Paris Student Exchange Program	Paris, France Spring 2017
Universidad de Chile B.S. in Engineering Sciences, Mathematics	Santiago, Chile 2014–2019
– Ranked in top 1%. 2 awards for outstanding academic achievement	

AWARDS & GRANTS

Fellowships & Grants:	
– Prob_AI Hub – Networking and Development Fund , £9,950 to organise the <i>Advances in Post-Bayesian Methods Workshop</i> .	2025
– Bloomberg Data Science Ph.D. Fellowship , 2 years (approx. \$180,000).	2024–Present
– UCL Departmental Teaching Assistant Studentship , 3.5 years (£150,000).	2022
– Center for Mathematical Modelling (CMM) MSc Studentship , approx. £10,000.	2020
– Selected Travel Grants: ICML 2023, Workshop on Functional Inference and Machine Intelligence 2024, BayesComp 2025. (approx. £1,500)	
Academic Prizes & Distinctions:	
– UCL Costas Goutis Prize , awarded for best PhD progress.	2024
– ICML Spotlight Presentation [2], top 3.5% of 9,000+ submissions.	2024
– Poster Prize , £500, judged by Prof. Guy Nason, UCL Departmental Poster Session.	2023
– Best MSc Thesis , Applied Mathematics, Universidad de Chile.	2022
– Best Paper Award , 10th International Conference on Methodologies and Intelligent Systems for Technology Enhanced Learning [7].	2020

TALKS & POSTERS

Approximately 15 talks at leading conferences/workshops/academic departments including:

- **Conferences & Workshops:** BayesComp Satellite Workshop 2025; International Conference on Robust Statistics (ICORS) 2025; Post-Bayes Seminar 2025; Bayesian Young Statisticians Meeting (BAYSM) 2025; ICML London Meetup 2024; UCL Distance-Based Methods in Machine Learning Workshop 2023.
- **Academic Departments & Industry:** Bloomberg L.P.; University of Cambridge, Department of Computer Science and Technology; Imperial College London, CSML Reading Group; Oxford-Man Institute Seminar; Newcastle University, Statistics Seminar; UCL Statistical Science PhD Seminars; UCL Department of Electronic and Electrical Engineering; Universidad Externado de Colombia; Universidad del Rosario.

SERVICES TO THE COMMUNITY

- **Reviewing Activities:**
 - Conference proceedings: AISTATS, NeurIPS, ICLR, ICML
 - Journals: JMLR, Statistics and Computing, Bayesian Analysis, TMLR
- **Organisation of Workshops & Seminars:**
 - Co-organiser of an AISTATS 2026 Workshop: *OPTIMAL: Optimisation and Post-Bayesian Inference in Machine Learning*, with confirmed keynote participation from Prof. Andrew Gordon Wilson and Prof. Pierre Alquier.
 - Co-organiser of the two-day *Advances in Post-Bayesian Methods Workshop* (2025) at UCL. Supported by over £18k in external funding, the event attracted **120+ attendees** from 12 countries, and featured eight invited talks (including Prof. Sonia Petrone and Prof. Peter Grünwald), 18 contributed talks (including Prof. Martyn Plummer and Prof. Yingzhen Li).
 - Co-organiser (with Dr. Jeremias Knoblauch, Prof. Pierre Alquier, and Dr. Edwin Fong) of the online *Post-Bayes Seminar Series* (ongoing). The series have featured invited speakers such as Prof. Dan Roy, Prof. Chris Holmes, and Prof. David Dunson.
 - Logistics Officer for the *UCL Distance-based Methods in Machine Learning Workshop* (2023)

PUBLISHED PAPERS

- [1] W. Laplante, **M. Altamirano**, A. Duncan, J. Knoblauch, and F.-X. Briol, “Robust and conjugate spatio-temporal Gaussian processes”, in *International Conference on Machine Learning*, 2025, pp. 32 562–32 592.
- [2] **M. Altamirano**, F.-X. Briol, and J. Knoblauch, “Robust and conjugate Gaussian process regression”, in *International Conference on Machine Learning*, spotlight, 2024, pp. 1155–1185.
- [3] G. Durán-Martín, **M. Altamirano**, A. Y. Briol Shestopaloff, L. Sánchez-Betancourt, J. Knoblauch, M. Jones, F.-X. Briol, and K. Murphy, “Outlier-robust Kalman filtering through generalised bayes”, in *International Conference on Machine Learning*, 2024, pp. 12 138–12 171.
- [4] **M. Altamirano**, F.-X. Briol, and J. Knoblauch, “Robust and scalable Bayesian online changepoint detection”, in *International Conference on Machine Learning*, 2023, pp. 642–663.
- [5] **M. Altamirano** and F. Tobar, “Nonstationary multi-output Gaussian processes via harmonizable spectral mixtures”, in *International Conference on Artificial Intelligence and Statistics*, 2022, pp. 3204–3218.
- [6] **M. Altamirano**, P. Uribe, D. Schlotterbeck, A. Jiménez, R. Araya, J. van der Molen Moris, and D. Caballero, “Unsupervised characterization of lessons according to temporal patterns of teacher talk via topic modeling”, *Neurocomputing*, 2021, ISSN: 0925-2312.

- [7] **M. Altamirano**, A. Jiménez, and R. Araya, “Lessons clustering using topics inferred by unsupervised modeling from textbooks”, in *International Conference in Methodologies and intelligent Systems for Technology Enhanced Learning*, Springer, 2020, pp. 85–94.

PREPRINTS

- [8] L. Rooijakkers and Joshua Rønneberg, F.-X. Briol, J. Knoblauch, and **M. Altamirano**, “Multi-output robust and conjugate Gaussian process regression”, Submitted to AISTATS 2026, 2025.
- [9] W. Laplante, **M. Altamirano**, A. Duncan, J. Knoblauch, and F.-X. Briol, “Conjugate generalised Bayesian inference for discrete doubly intractable problems”, To be submitted to JASA, 2025.

TEACHING & SUPERVISION

Teaching

- **Teaching Assistant**, University College London (UCL) 2023–present
Introduction to Practical Statistics (Term 1 2022 / Term 2 2022 / Term 2 2023), Computing for Practical Statistics (Term 2 2022), Programming Fundamentals (Term 1 2023 / Term 1 2024 / Term 1 2025)
- **Teaching Assistant**, Universidad de Chile 2016–2020
Multivariable Calculus (Fall 2016 / Fall 2018 / Fall 2019 / Spring 2019 / Fall 2020), Advanced Calculus and Applications (Spring 2019), Linear Algebra (Spring 2017 / Spring 2018), Introduction to Calculus (Spring 2017)

Supervision

- Joshua Rooijakkers, MSc Econometrics & Management Science at Erasmus University Rotterdam. 2025
Supervised MSc thesis: “Robust and Conjugate Multi-Output Gaussian Processes”. The thesis was later developed into a paper submitted to AISTATS 2026.

EXPERIENCE

Bloomberg L.P. Research Intern	London, UK July 2025–September 2025
Matterhorn Studio Research Scientist	London, UK January 2024–September 2024
Universidad de Chile - Center for Mathematical Modeling Research Assistant, Supervisor: Dr. Joaquín Fontbona	Santiago, Chile September 2021–June 2022
Universidad de Chile - Center for Advanced Research in Education Research Assistant, Supervisor: Dr. Roberto Araya	Santiago, Chile March 2019–January 2022
– Designed and implemented unsupervised learning models to analyse temporal patterns of teacher talk. – Published findings in reputable journals and conferences as an undergraduate: [1], [2]	

CODING AND COMPUTER SKILLS

- **Languages:** Python, Matlab, Java, SQL, R
- **Additional Skills:** Linux, L^AT_EX, Github

LANGUAGES

- **Spanish:** Native Speaker
- **English:** Fluent
- **French:** Basic